

	(b)	Any 4 × (1) 1) Counts is reduced significantly (or equivalent alternatives e.g. by almost a half) by the paper, so alpha particles present (✓) 2) Another significant reduction (or alternative e.g. essentially all of the remaining radiation is stopped) by the aluminium, so beta particles present (✓) 3) Count with lead is larger than with aluminium (but almost the same) so no gamma present (✓) 4) The measured counts with aluminium and lead are essentially the same, so this is because of background radiation / the background radiation is approximately 25 counts per minute. (✓) 5) Randomness of nuclear decay is the reason for increased value with lead. (✓)		4		4		4
		Question 5 total	0	8	0	8	4	4